Empowering Small Businesses with Tech:

Digital Record-Keeping for Financial Best Practices and Business Growth.

Summary of Challenges and Opportunities.

Introduction

Small firms are the backbone of job creation in developing countries. Widespread reliance on paper records among them undermines precision, analytics, and the potential for business partnerships.

Small businesses in Nigeria and Sub-Saharan Africa face significant opportunities for growth and job creation. However, many struggle with informed decision-making, business growth, loan access, and forming partnerships due to a lack of crucial business and financial information. Our extensive data from three pilot studies conducted since 2019 consistently reveals that over 70-90% of businesses in more than 100 markets in Lagos still rely on paper records. Even in bustling urban areas, these businesses adhere to paper methods, citing time, attention, and mental effort involved in using new technologies. However, this widespread practice lacks organization, accuracy, and precision, rendering it unsuitable for effective data analysis. This limitation may impede precise decision-making necessary for growth and the establishment of successful business partnerships.

Our research has two main goals. First, we examine how businesses are impacted as they transition from using paper to employing a simple mobile app for record-keeping. We observe how this change affects their business performance, access to loans, and the utilization of data. Second, we study the impact of providing micro-consulting services alongside meticulous record-keeping. We employ both randomized control trials and conduct in-depth interviews with Lagos market vendors for our approach. In this document (summary of findings), while the outcome of digital record-keeping is still ongoing, we emphasize the challenges in technology adoption and the opportunities.

Challenges

Despite the promising tech readiness of small firms in Lagos, there persists low motivation to adopt new technology, fueled by training gaps and a lack of internal expertise among small firms.

1. The Fogg Behavioral Model, crafted by DJ Fogg at Stanford Behavior Design Lab, sheds light on why small businesses in Lagos might be reluctant to embrace technology. According to Fogg, businesses require three essential elements to adopt new behaviour: a convincing motivation, a user-friendly tool, and a timely prompt "reminder" to incorporate technology into their daily operations.

Drawing on Fogg's concept of the necessity for firms to be motivated to change behavior, our research indicates that small businesses in Lagos exhibit minimal motivation to use technology for record-keeping. This lack of motivation is frequently attributed to low or inconsistent sales, a limited customer base, and a weak understanding of the overall benefits of record-keeping.

Fogg identified that the "fear or hope" in technology is the driving force for adoption. We find partial support for both the fear of non-adoption and the hope and benefits of adoption. Regarding "hope," more than half of the firms (54 percent) that keep records either perceive apps as less necessary than paper forms or are unsure about their utility for their business, and approximately 40 percent of the total respondents don't maintain any records at all.

This implies that only about 46 percent, or roughly half of the potential market, see promise in using an app. In terms of "fear" for non-adoption, our research suggests that only a few firms are interested in using digital apps due to concerns about keeping their records insecure.

It remains uncertain whether instilling hope or signaling the fear of not keeping records is an effective way to encourage technology use. This uncertainty leaves us unsure about whether firms have much to lose or gain by not embracing technology as we test the effect of certain interventions.

However, while it remains unclear whether firms want to use apps, as the numbers indicating usefulness or not are often evenly divided, what is evident is that small firms may have a limited capacity to utilize business-related apps. Despite the widespread use of technology, such as mobilephones (98.4%), for various activities like calls, messages, chatting, and app usage, a substantial portion (78.7%) of these businesses has not received any training, and many firms do not maintain good record-keeping practices (see Fig 1).

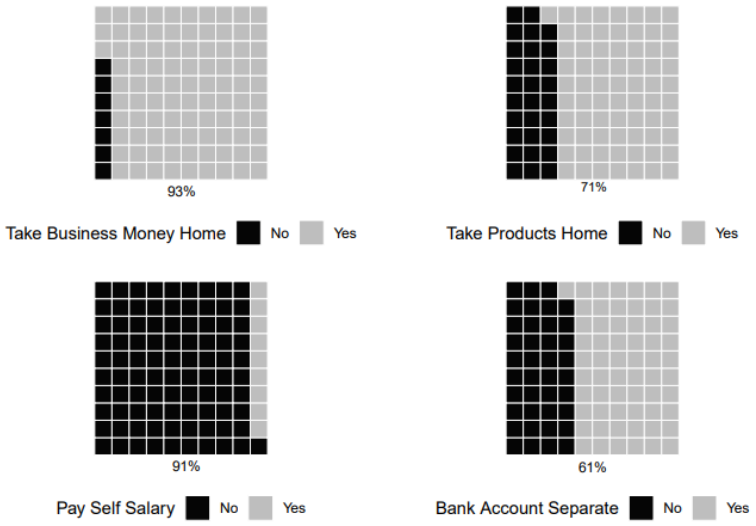

Fig 1: Many small firms do not maintain good record-keeping practices, including issues such as not paying oneself a salary or neglecting to record business products taken home:

In settings with limited infrastructure and infrequent internal changes in firm structure, like informal markets, owners rely more on personal memory than institutional memory for decision-making, though both methods are embraced.

2. Our research revealed that the majority of small businesses (65.9%) prioritize memory over written records (52%) when making decisions, though many employ a combination of both methods. Interestingly, businesses existing for over nine years are even less inclined to maintain records (see Fig 2) indicating some mastery over their business patterns.

In addition to mastering business patterns, which may be commonplace among long-existing firms in informal markets who may also be less likely to scale and experience business changes over time, a significant factor likely influencing the reliance on personal memory is the possibility that firms may not want to fully depend on mobile tech tools, especially in situations where there is a potential for the loss or low battery of mobile phones.

This circumstance may hinder real-time record-keeping and disrupt the process for these businesses. These challenges are likely prevalent among small businesses in developing countries grappling with infrastructural issues.

Fig 2: After nine years of business experience, we start to notice a lower proportion of record-keepers compared to non-keepers.

Triggers (training and reminders) work, but their effectiveness depends on minimizing the user-related burden cost) associated with using the tech tool. Additionally, if the costs are too low or too high, adoption decreases.

3. Fogg's model suggests that triggers, those things prompting action are critical in changing behavior. To encourage businesses to use an app, we employed various triggers like training and reminders. However, their effectiveness primarily relies on how prepared businesses are and their perception of associated costs.

Speaking of costs, crucial especially for small businesses, there are four types to consider. Heuristic cost involves the effort or thinking businesses need to put in, varying from easy to challenging. Attentional cost pertains to how much attention businesses need to give while using digital tools for real-time data collection.

Informational cost delves into the perceived value businesses see in using the app, questioning its worth for their operations. Lastly, Intellectual cost reflects how much businesses need to use their brains (including learning curve cost), with some finding it straightforward and others a bit challenging.

4. Small businesses that sell similar products and don't have much inventory will have low or no extra costs. While it is important to reduce the burden, firms who are unwilling to bear no cost mainly often drop out easily. This means they might not see the need for systematic record-keeping.

Moreover, when businesses are burdened with multiple costs, they may recognize

Digital technologies are not fully tailored to specific needs of small firms resulting in a high dropout.

Following an extensive assessment of available e-bookkeeping apps in Nigeria, we found that the e-bookkeeping market, which emerged only in the past five years, comprises notable applications. However, even with robust mobile applications, small businesses in informal settings face additional challenges. For instance, as previously mentioned, firms may be reluctant to use an app if they fear that any issues with their phones could disrupt their business operations. We are at a crossroads where it is crucial to develop record-keeping technology beyond the basic function of providing patterns, incorporating traditional methods of record-keeping such as image recognition from paper records, and expanding beyond efficient record-keeping.

Many apps struggle with stock management, integration with other business apps, and business analytics.

In our ongoing experimental pilot research, we are still assessing whether employing digital methods is more effective than relying on paper or personal memory, which we believe it should be. Continuous use of paper records might result in mistakes and the incorrect recall of information over time (recall bias). However, to be certain, we need extensive research involving more businesses using a record-keeping app to fully understand its implications.

Tricky Leverages

Loan requirements vary. As digital records are not universally mandated for loans, addressing challenges and capitalizing on opportunities in digital record-keeping adoption requires a nuanced approach including incentives for early adopters.

Using digital records to secure loans is not a common approach. This year, our data shows that a significant number of businesses (57.7%, 946 respondents) don't even need a loan and only about 4.5% mentioned they benefit from records because it helps in securing a loan. However, for the 43% who do receive a loan, we believe that maintaining digital records is a useful means to help secure business partnerships including access to loans.

Firms that have never applied for loans usually cite high-interest rates, and organizations offering lower rates, such as Lagos State Employment Trust Fund (LSETF), can be highly competitive. According to our 2023 data, about 72-85% (2019, 2022 data) of those who obtained a loan did so from either microfinance banks or commercial banks. The network of small firms benefiting from adopting digital record-keeping might expand with the availability of low-interest loans and additional incentives for maintaining digital records. This incentive includes provisions for low-interest loans to purchase digital assets for initial acquisition or in case of loss.

The Opportunities

Despite the challenges of technology adoption, in-depth interviews with firm owners reveal that, compared to paper records, record-keeping app makes the process more accessible and efficient.

Digital record-keeping enables small firms to have their records accessible on the go, making them available anytime and anywhere. It enhances reminders and notifications, especially for stocks. It provides quicker access, minimizes the risk of loss from events like fire or natural disasters, and facilitates efficient sorting and filtering through business activities and time.

Digital technology that incorporates traditional methods is likely to encourage adoption.

1. Most small business owners (72%) think an app that lets them take pictures of their paper records and organizes them automatically will make them more likely to use it for managing their records. This means that new technology for small businesses can integrate tools from artificial intelligence and deep learning.

There is a demand for accurate and precise records especially in stock management , which paper-records are less likely to provide.

2. Our research shows that small businesses , who keep records, want accurate records, especially with almost 90% wanting real-time updates on their stock and around 81% desiring precise knowledge of their profits, or at least a value

very close to the exact amount. In fact, about 7 in 10 business owners believe that training (which may include better practices such as tech) on record-keeping is necessary for running their business.

3. Our research shows that stock management is the most important reason small firms may need record keeping. As it stands from our assessment of e-bookkeeping apps in Nigeria, that functionality still lags behind.

There is potential in utilizing early adopters as peer mentors. Future research and tech development should explore peer mentoring.

4. Looking at how people who started using it early (early adopters) help others could be a topic for more research and improvement. For instance, our research found that a small but important number (15.96%) ask their colleagues for help when making strategic business decisions, suggesting there could be potential for support and mentoring from peers.

Small business profile: On average, they're 36 years old with limited formal education (57% completed secondary school, 34% have a diploma). Among them, 58% are women, and they operate in less upscale commercial areas, making around N20,000 in sales daily. In 2022/2023, we surveyed 1,928 firms, compared to 277 in 2020 and 432 in 2019. Our respondents include vendors selling various products, covering more than 40 different types.

Research Reach in Lagos in 2022.

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The research and project implementation are currently led by Oluwayomi Akosile, Omotola Ogunlana is the Training Manager and co-assist research implementation. The Principal Investigator is Abiola Oyebanjo. Previous projects and research were led by Adebisi Nofiu, Omolola Sameen, Olowe Medinat, and Adeniyi Festus.

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